# DATA GOVERNANCE AND SECURITY DASHBOARD

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Role :- Business Analyst Project

# PROJECT OVERVIEW :-

- This project focuses on building an interactive Data Governance and ESG Dashboard to analyze environmental performance across countries and regional aggregates using World Bank data. The objective is to visualize and compare key ESG indicators—particularly $CO_2$ emissions per capita—through map visualizations, bar charts, KPI cards, and interactive filters for ESG categories (Environment, Social, Governance) and Country Codes.

# DATASET OVERVIEW :-

- The dataset consists of longitudinal country-level data containing 16,000+ rows and 60+ columns, including Country Name, Country Code, Indicator Name, Indicator Code, and yearly values from 1960 to 2020 (with 2019 used for analysis due to data availability). It includes both real countries and regional aggregates such as ARB, CSS, and EMU. The dashboard enables users to filter by indicator, category, and country, while tooltips provide detailed series descriptions and ESG values, supporting data-driven insights into global emission patterns and regional disparities.

## ESG Category

Environment

Social

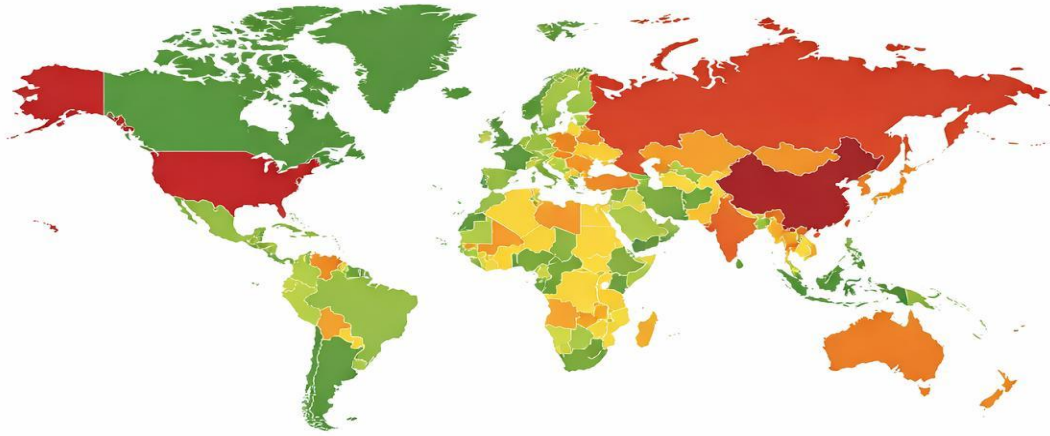
Governance

**Select Indicator**

CO2 Emissions

## ESG Indicator by Country

Year: 2020



# INSIGHTS FROM ESG INDICATOR BY COUNTRY (HEAT MAP CHART) :-

- North America and Russia show high emission intensity . The United States and Russia appear in deep red shades, indicating high per-capita $CO_2$ emissions compared to most other regions.
- China shows significantly high emissions . China is highlighted in dark red, reflecting its large industrial base and coal-dependent energy structure.
- Europe demonstrates moderate emission levels . Most European countries fall in yellow to light orange shades, suggesting relatively controlled emissions compared to major industrial emitters.
- Africa has predominantly low emissions . Many African countries are in green shades, indicating low per-capita emissions, largely due to lower industrialization and energy consumption levels.
- Clear global disparity in emission distribution . The map highlights a strong contrast between highly industrialized economies (red/orange) and developing regions (green), showing uneven environmental impact globally.

## ESG Category

Environment

Social

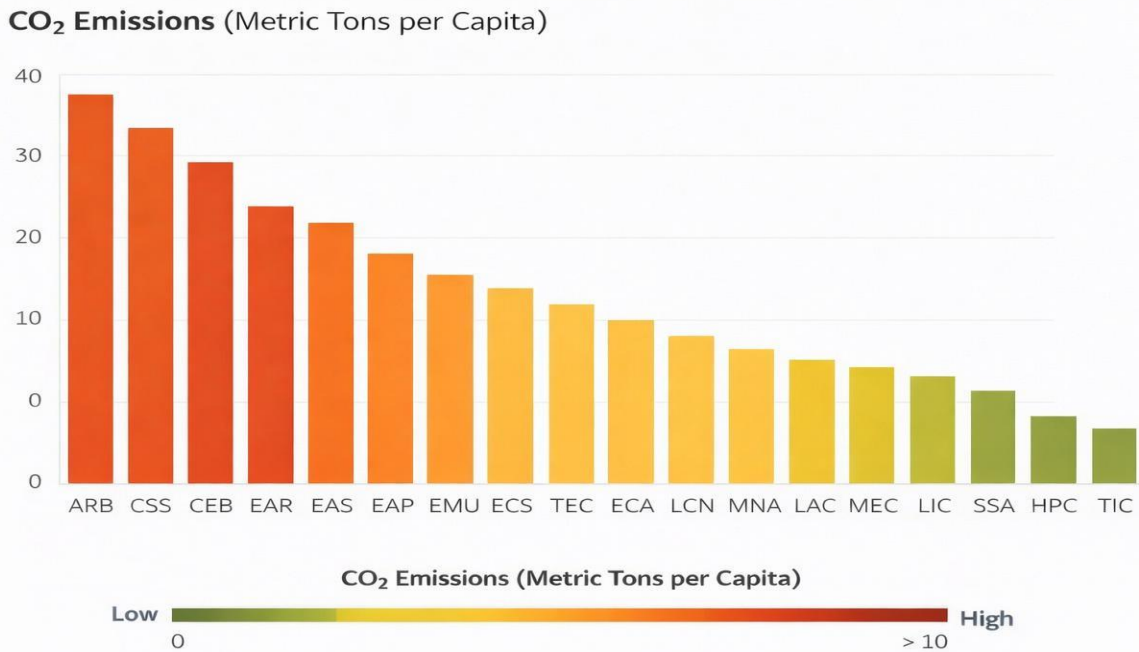
Governance

**Select Indicator**

$CO_2$ Emissions

## $CO_2$ Emissions by Country – 2020

**Sort by:** Descending



# INSIGHTS FROM CO2 EMISSIONS BY COUNTRY (BAR CHART) :-

- Arab World (ARB) shows the highest emissions . ARB has the highest $CO_2$ emissions per capita among the listed groups, indicating energy-intensive economies and high fossil fuel dependency in that region.
- Small States (CSS) and Central Europe & Baltics (CEB) are also high emitters . These regions appear in the top range, suggesting relatively higher industrial activity or energy consumption per capita.
- European Union (EMU/ECS) falls in the mid-range . European regions are not at the top but still show moderate emission levels, reflecting industrial presence along with environmental regulations.
- Low Income Countries (LIC) and Sub-Saharan Africa (SSA) show very low emissions . These regions have the lowest per-capita $CO_2$ emissions, which typically correlates with lower industrialization and energy usage.
- Clear emission gap between high-income and low-income regions . There is a strong visible gradient from ARB (high) to LIC/SSA (low), indicating economic development level significantly influences per-capita emissions.

## ESG Category

Environment

Social

Governance

**Select Series Code**

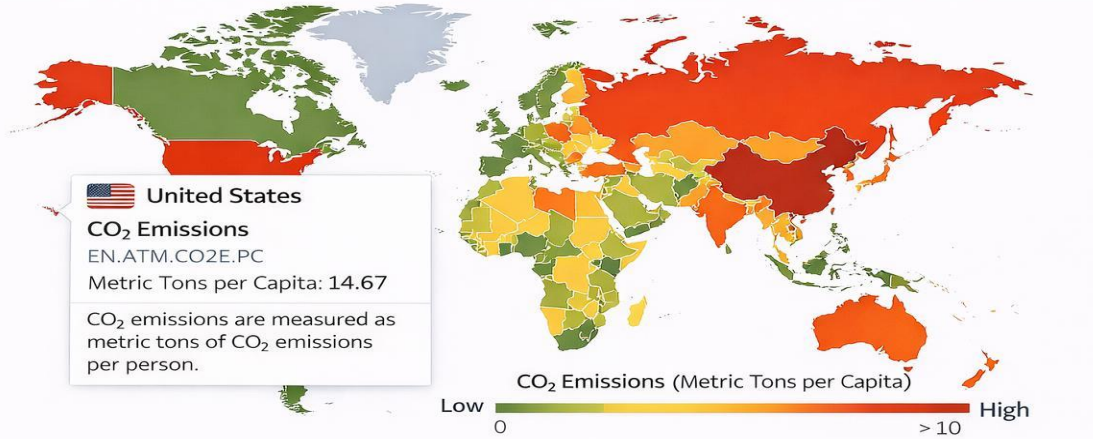
EN.ATM.CO2E.PC

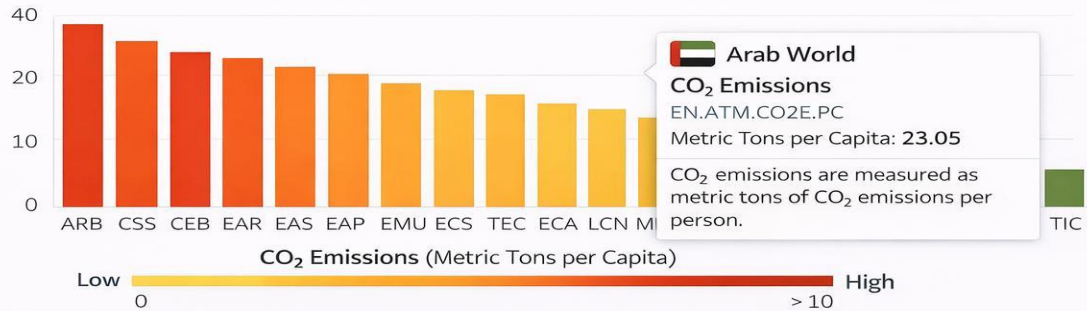
**Country Code**

ARB

## $CO_2$ Emissions by Country

2020



# KPI CARDS :-

| KPI's | Values |
|---|---|
| Global Electricity Access | 89.6 % |
| Global Life Expectancy | 72.6 years |
| Global CO2 Emissions | 4.56 metric tons per capita |
| LMIC Unemployment rate | 6.3 % |
| Education Gender Parity | Index – 0.99 |
| Avg GDP Growth (historical) | 3.7 % |

# PROJECT CONCLUSION :-

- This ESG Dashboard project successfully demonstrates how data visualization and governance principles can be applied to analyze global environmental performance using structured country-level data. By integrating interactive maps, bar charts, KPI cards, and dynamic filters, the dashboard enables users to explore $CO_2$ emissions across countries and regions in a clear and meaningful way. The analysis highlights significant disparities in per-capita emissions between developed and developing economies, emphasizing the relationship between industrialization and environmental impact.From a Business Analyst perspective, the project showcases skills in data cleaning, transformation, indicator selection, handling aggregates vs. country-level data, and designing user-focused interactive dashboards. Overall, the solution supports data-driven decision-making by transforming raw longitudinal ESG data into actionable insights through structured visualization and interactivity.

Arab World

$CO_2$ Emissions

EN.ATM.CO2E.PC

Metric Tons per Capita: 23.05

$CO_2$ emissions are measured as metric tons of $CO_2$ emissions per person.

# Data Governance & ESG Dashboard

| Global Electricity Access | Global Life Expectancy | Global $CO_2$ Emissions | LMIC Unemployment | Avg GDP Growth |
|---|---|---|---|---|
| 89.6 % | 72.6 years | 4.56 metric tons per capita | 6.3 % | 3.7 % |

**Select Indicator**

$CO_2$ Emissions

**Year**

2020

**ESG Category**

Environment

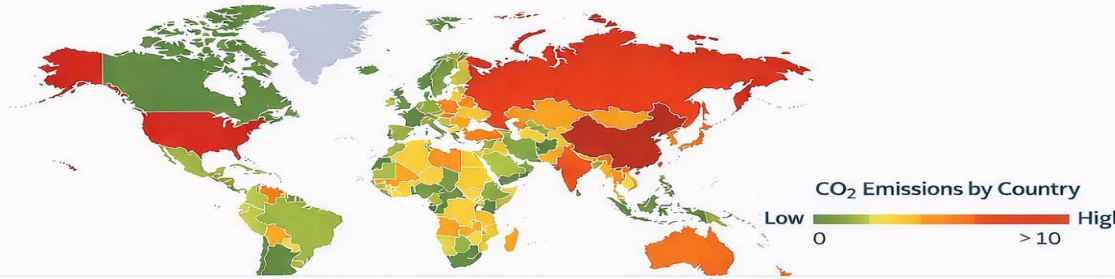
Social

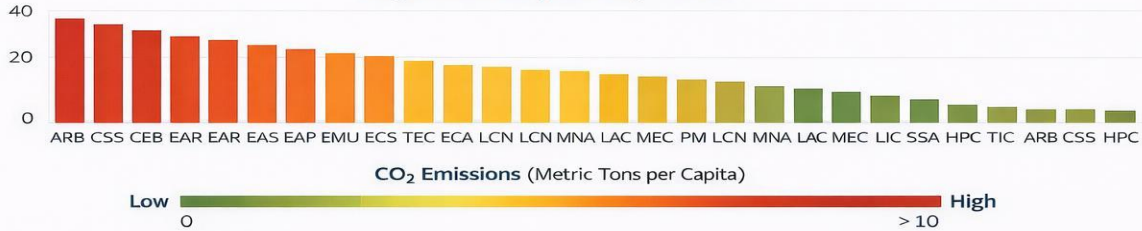
Governance

**Education Gender Parity**

Index – 0.99

## ESG Indicator by Country




## $CO_2$ Emissions by Country – 2020